

AARHUS UNIVERSITY

DEPARTMENT OF NERDS (MATH)

INTRODUCTION TO ALGEBRA 1

If this is the title, please don't read
me...

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Goofy ah Math is scawy oWo

Exercise: In the following, n denotes a natural number.

1. Show, by induction on n , that

$$\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

The base case for $n = 1$

$$\text{LHS: } \frac{1}{1^2} = 1 \quad \text{RHS: } 2 - 1 = 1$$

the base case is satisfied since $1 \leq 1$

Next step is the hypothesis for induction ThingyMaBob, which is to assume it holds for some positive number $\Omega = n$ (Queue introduction to the Omegaverse) #KnotMeDaddyFunch

The Goofy ah Ω goes to town #Purr

$$\sum_{k=1}^{\Omega} \frac{1}{k^2} \leq 2 - \frac{1}{\Omega}$$

Now it is time for marriage, which is to prove it for $\Omega + 1$, the next number in the chain, so the actual induction step.

But remember as a true Alpha, we know the left side is just the summation:

Goofy ah søren, insert summation or something

2. Conclude that for all natural numbers n , it holds that

$$\sum_{k=1}^n \frac{1}{k^2} < 2.$$

3. Consider whether the conclusion in 2 can be shown directly by induction on n .